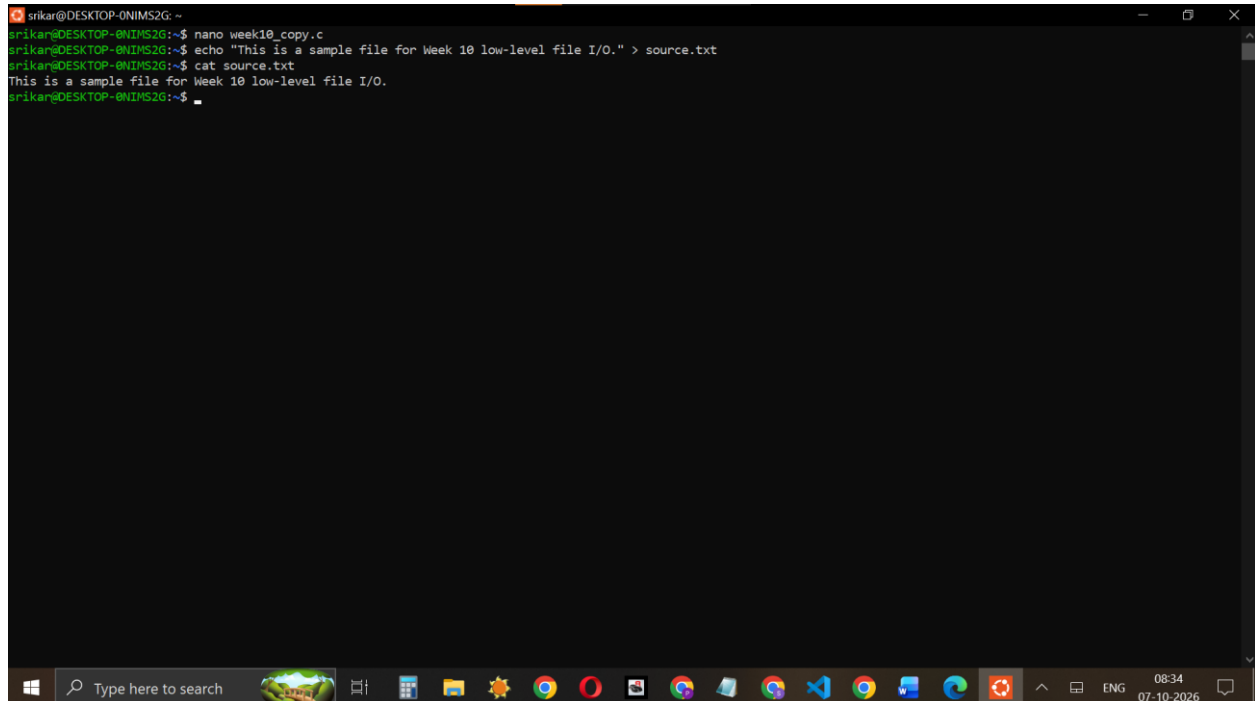


Practical Week-10

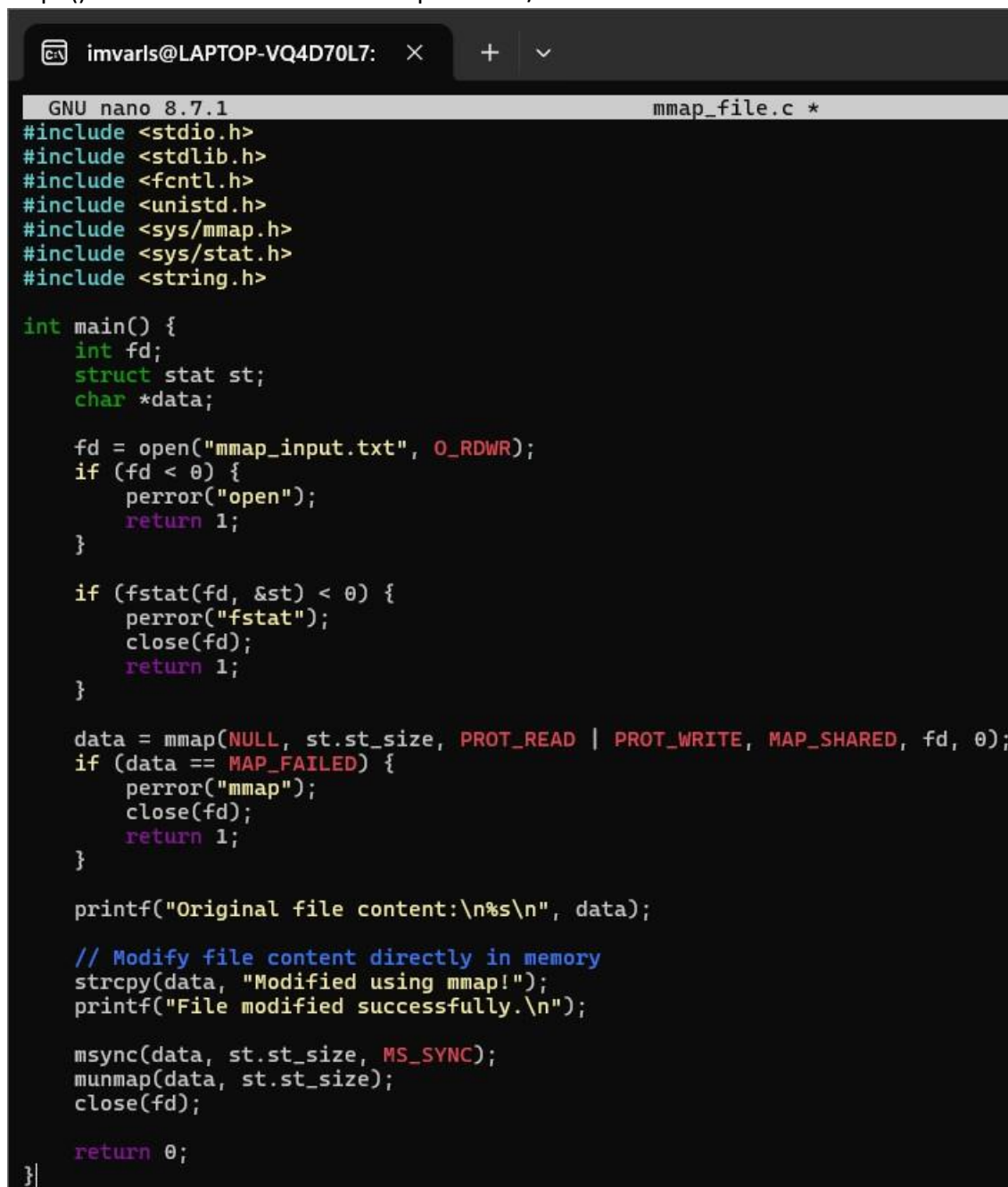
1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).



```
srikar@DESKTOP-0NIMS2G: ~  
srikar@DESKTOP-0NIMS2G:~$ nano week10_copy.c  
srikar@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 10 low-level file I/O." > source.txt  
srikar@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 10 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$
```

```
srikar@DESKTOP-0NIMS2G:~$ echo "This is a sample file for Week 10 low-level file I/O." > source.txt  
srikar@DESKTOP-0NIMS2G:~$ cat source.txt  
This is a sample file for Week 10 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ gcc week10_copy.c -o week10_copy  
srikar@DESKTOP-0NIMS2G:~$ ./week10_copy  
File copied successfully using low-level system calls.  
Current file position: 54 bytes  
srikar@DESKTOP-0NIMS2G:~$ nano week10_standard.c  
srikar@DESKTOP-0NIMS2G:~$ gcc week10_standard.c -o week10_standard  
srikar@DESKTOP-0NIMS2G:~$ ./week10_standard  
File copied successfully using standard library functions.  
srikar@DESKTOP-0NIMS2G:~$ cat copy_week10_standard.txt  
This is a sample file for Week 10 low-level file I/O.  
srikar@DESKTOP-0NIMS2G:~$ time ./week10_copy  
File copied successfully using low-level system calls.  
Current file position: 54 bytes  
  
real    0m0.002s  
user    0m0.000s  
sys     0m0.002s  
srikar@DESKTOP-0NIMS2G:~$ time ./week10_standard  
File copied successfully using standard library functions.  
  
real    0m0.016s  
user    0m0.000s  
sys     0m0.007s  
srikar@DESKTOP-0NIMS2G:~$
```

2. Develop a program that redirects standard input and output using file descriptors and `dup2()`. Demonstrate how shells implement I/O redirection.



```
GNU nano 8.7.1 mmap_file.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mmap.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    int fd;
    struct stat st;
    char *data;

    fd = open("mmap_input.txt", O_RDWR);
    if (fd < 0) {
        perror("open");
        return 1;
    }

    if (fstat(fd, &st) < 0) {
        perror("fstat");
        close(fd);
        return 1;
    }

    data = mmap(NULL, st.st_size, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
    if (data == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    printf("Original file content:\n%s\n", data);

    // Modify file content directly in memory
    strcpy(data, "Modified using mmap!");
    printf("File modified successfully.\n");

    msync(data, st.st_size, MS_SYNC);
    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

Output:

```
srikar@DESKTOP-0NIMS2G: ~  
real    0m0.016s  
user    0m0.000s  
sys     0m0.007s  
srikar@DESKTOP-0NIMS2G:~$ nano week10_redirect.c  
srikar@DESKTOP-0NIMS2G:~$ gcc week10_redirect.c -o week10_redirect  
srikar@DESKTOP-0NIMS2G:~$ echo "This is input redirection using dup2 in Week 10." > input_week10.txt  
srikar@DESKTOP-0NIMS2G:~$ cat input_week10.txt  
This is input redirection using dup2 in Week 10.  
srikar@DESKTOP-0NIMS2G:~$ ./week10_redirect  
srikar@DESKTOP-0NIMS2G:~$ cat output_week10.txt  
This is input redirection using dup2 in Week 10.  
srikar@DESKTOP-0NIMS2G:~$ ls -l week10_copy.c week10_standard.c week10_redirect.c source.txt copy_week10_lowlevel.txt copy_week10_standard.txt input_week10.txt output_week10.txt  
-rw-r--r-- 1 srikar srikar  54 Oct  7 03:08 copy_week10_lowlevel.txt  
-rw-r--r-- 1 srikar srikar  54 Oct  7 03:09 copy_week10_standard.txt  
-rw-r--r-- 1 srikar srikar  49 Oct  7 03:10 input_week10.txt  
-rw-r--r-- 1 srikar srikar  49 Oct  7 03:11 output_week10.txt  
-rw-r--r-- 1 srikar srikar  54 Oct  7 03:03 source.txt  
-rw-r--r-- 1 srikar srikar 1180 Oct  7 03:03 week10_copy.c  
-rw-r--r-- 1 srikar srikar  832 Oct  7 03:10 week10_redirect.c  
-rw-r--r-- 1 srikar srikar  751 Oct  7 03:06 week10_standard.c  
srikar@DESKTOP-0NIMS2G:~$
```

Windows taskbar at the bottom shows the search bar, task view, and various application icons. The system tray on the right displays the time as 08:43 and the date as 07-10-2026.